

Suppressing Lithium Dendrite Growth in Argyrodite Sulfide Electrolytes with Fluorinated Polymer Interlayers

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Abstract

All-solid-state batteries that pair a lithium metal anode with a sulfide electrolyte promise high energy density, but lithium filaments penetrate the electrolyte at modest current densities and short-circuit the cell. Here we show that a 40 nm interlayer of a fluorinated polymer, cast from solution onto Li₆PS₅Cl pellets, raises the critical current density from 0.6 to 2.4 mA cm⁻² at 25 °C. Cryogenic electron microscopy reveals that the interlayer decomposes into a LiF-rich interphase that is electronically insulating yet conducts lithium ions, which homogenises plating. Symmetric cells cycled for 1,200 hours at 0.5 mA cm⁻² without short circuits, and full cells with a nickel-rich cathode retained 89% of their capacity after 500 cycles. The results indicate that interface chemistry, rather than electrolyte density alone, governs dendrite initiation in argyrodite systems.

1. Introduction

Replacing the graphite anode of a lithium-ion cell with lithium metal could raise its specific energy by roughly 40%. Sulfide electrolytes of the argyrodite family are attractive partners because their room-temperature ionic conductivity exceeds 3 mS cm⁻¹ and they can be densified by cold pressing. Their weakness is the interface: lithium deposits preferentially at grain boundaries and voids, and the resulting filaments grow through the pellet within a few cycles once a critical current density is exceeded.

Previous strategies have targeted the bulk electrolyte, for example by hot pressing to reduce porosity, or the anode, for example by alloying lithium with indium. Both reduce the achievable energy density. We instead ask whether a nanometre-scale artificial interphase can redistribute the

current at the anode surface without adding measurable mass.

2. Methods

Li6PS5Cl was synthesised by ball milling stoichiometric Li2S, P2S5 and LiCl for 20 hours, followed by annealing at 550 °C under argon. Pellets of 10 mm diameter were cold pressed at 375 MPa. The interlayer was deposited by spin coating a 0.5 wt% solution of a poly(vinylidene fluoride-co-hexafluoropropylene) derivative in dimethyl carbonate. Symmetric Li|Li6PS5Cl|Li cells were cycled under a stack pressure of 5 MPa. Critical current density was determined by increasing the current in steps of 0.1 mA cm-2 every two cycles until a voltage drop indicated a short circuit.

3. Results

Cell	Interlayer thickness (nm)	Critical current density (mA cm-2)	Hours to short circuit at 0.5 mA cm-2
Bare electrolyte	0	0.6	46
Thin coating	15	1.3	310
Optimised coating	40	2.4	> 1,200
Thick coating	120	1.9	> 1,200 (high overpotential)

The optimised coating raised the critical current density fourfold. Thicker coatings also prevented shorting but increased the interfacial resistance from 12 to 85 ohm cm2. Microscopy of cycled cells showed a continuous 8 nm LiF-rich layer with no lithium penetration into grain boundaries.

4. Discussion

Our phase-field simulations reproduce the observed trend when the interphase is modelled as an electronic insulator with an ionic conductivity of 10-6 S cm-1, suggesting that suppressing electron leakage, not mechanical blocking, is the dominant mechanism. Scaling the coating to pouch cells is the subject of ongoing work.

References

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